

❖ Section 5. Stationary Distribution and Convergence

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Stationary Distribution

Irreducible and regular

Perron-Frobenius Theorem

Stationary distribution:

Let Ω be a state space, $\Omega = \{1, 2, 3, \dots, m\}$.

$$X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_{n-1} \rightarrow X_n \rightarrow X_{n+1}, \dots$$

A probability distribution $\pi_* = (\pi_i)_{i \in \Omega} = (\pi_1, \dots, \pi_m)$ on Ω is **stationary** if it remains unchanged in the Markov chain as time progresses.

$$\pi_* = \pi_* P$$

$$[\pi_1, \dots, \pi_m] = [\pi_1, \dots, \pi_m] \begin{bmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{bmatrix}$$

For each state j ,

$$\pi_j = \sum_i \pi_i p_{ij} \quad \text{for all states } j \in \Omega$$

Linear Algebra Point of View:

That is $\vec{w}^T = \pi_*$ is a **left-eigenvector** of P with **eigenvalue 1**.

Take transpose, $\vec{w}$ is the eigenvector of P^T with eigenvalue 1.

$$P^T \vec{w} = \vec{w}$$

$$\vec{w} = \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_n \end{bmatrix}$$

Eigenvalue 1 always exists, since rows of P sum to 1:

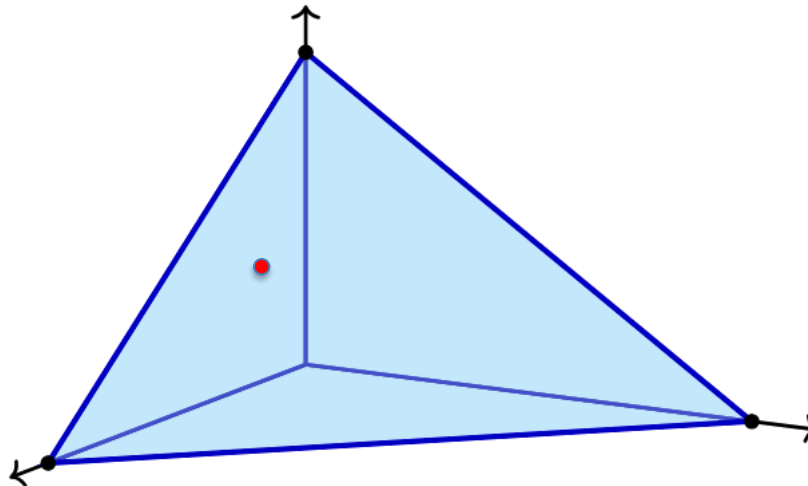
$$\begin{bmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

From linear algebra, P and P^T have the same non-zero eigenvalues.

We regard the Markov chain as a map on probability distributions, via the mapping

$$F: \mathbb{R}^m \rightarrow \mathbb{R}^m \quad F(\mu) = \mu P \quad \text{or} \quad F: \mu_j \rightarrow \sum_i \mu_i p_{ij}$$

For example, A probability distribution on $\Omega = \{1,2,3\}$ is a triple $\mu = (\mu_1, \mu_2, \mu_3)$ such that $\mu_i \geq 0$ and $\mu_1 + \mu_2 + \mu_3 = 1$



The stationary distribution is a **fixed point** for this map F .

$$F(\pi_*) = \pi_*$$

Lemma: Suppose π_* is a stationary distribution and $P(X_0 = j) = \pi_j$ for all states $j \in \Omega$. Then,

$$P(X_n = j) = \pi_j \text{ for all states } j \in \Omega \text{ and all } n \geq 1.$$

Proof:

$$\begin{aligned} P(X_1 = j) &= \sum_i P(X_1 = j | X_0 = i) P(X_0 = i) \\ &= \sum_i p_{ij} \pi_i \\ &= \pi_j \end{aligned}$$

Then repeat the argument for X_2 by conditioning on X_1 , and so on.

Reason for calling a stationary distribution: Once the chain enters this distribution it must remain there.

Stationary does **not** mean “the chain stops moving”.

It means the **distribution** does not change

Several immediate questions:

- Does it always **exist** a stationary distribution?
- If it exists, is the stationary distribution **unique**?
- Does the chain always **converge** to the stationary distribution?

In general, the answer is `no'. However, under some assumptions,

- Every **finite** Markov chain has **at least one** stationary distribution.
- If the chain is **irreducible**, then there exists a **unique** stationary distribution.

Example (Weather chain)

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$

Direct equation:

$$(\pi_1, \pi_2) = (\pi_1, \pi_2) \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \quad \begin{cases} \pi_1 = 0.8\pi_1 + 0.4\pi_2 \\ \pi_2 = 0.2\pi_1 + 0.6\pi_2 \\ \pi_1 + \pi_2 = 1 \end{cases}$$

Transposed equation (recommended)

Solve $P^T \vec{w} = \vec{w}$. Equivalently, $(P^T - I)\vec{w} = \vec{0}$

$$\pi_* = \left(\frac{2}{3}, \frac{1}{3} \right)$$

Example: Balls in boxes

The transition matrix is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$

Solve $P^T \vec{w} = \vec{w}$, by

$$(P^T - I)\vec{w} = \vec{0}$$

Find the eigenvectors corresponding to eigenvalue 1,

$$\pi_* = \left(\frac{1}{6}, \frac{2}{3}, \frac{1}{6} \right)$$

Note: MATLAB and Python use linear algebra convention for computation.

□ Irreducible and regular

Let P be the transition matrix of a Markov chain.

Definition:

(1) The Markov chain is **irreducible** if for each states i, j , there is an integer n such that $p_{ij}^{(n)} = (P^n)_{ij} > 0$.

(2) The Markov chain is **regular** if there is an integer n such that P^n is strictly positive, i.e., $(P^n)_{ij} > 0$ for all i, j .

Remark:

For any state of the irreducible Markov chain, there is a positive probability of visiting all other states.

Regular implies irreducible. The converse is not true.

Example 1

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/8 & 3/4 & 1/8 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

Example 2

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Note:

Let P be a stochastic matrix with dimension d .

- If P is **regular**, then P^d is strictly positive.
- If P is **irreducible**, then $P + P^2 + P^3 + \dots + P^d$ is strictly positive.

□ Perron-Frobenius Theorem.

Theorem [Perron-Frobenius] Suppose P is a *regular* $n \times n$ transition matrix. Then there is a unique, strictly positive probability vector $\vec{w} = \boldsymbol{\pi}_*^T \in \mathbb{R}^n$ such that

$$\pi_* P = \pi_*$$

and such that

$$\lim_{k \rightarrow \infty} P^k = W := \begin{bmatrix} \pi_1 & \pi_2 & \cdots & \pi_n \\ \pi_1 & \pi_2 & \cdots & \pi_n \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \cdots & \pi_n \end{bmatrix} = \begin{bmatrix} -\vec{w}^T & - \\ -\vec{w}^T & - \\ \vdots & \vdots \\ -\vec{w}^T & - \end{bmatrix} = \vec{e} \vec{w}^T$$

$$\text{Here } \vec{e} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad \vec{w} = \begin{bmatrix} \pi_1 \\ \vdots \\ \pi_n \end{bmatrix}$$

Proof: See MATH 5110 lecture notes.

Corollary: For any probability distribution $\mu = \vec{v}^T$, we have $\lim_{k \rightarrow \infty} \mu P^k = \pi_*$.

Since, $\lim_{k \rightarrow \infty} P^k = \vec{e} \vec{w}^T$, we have $\lim_{k \rightarrow \infty} \vec{v}^T P^k = v^T \vec{e} \vec{w}^T = \vec{w}^T = \pi_*$

This means that no matter what probability distribution we start, the limit of $\vec{v}^T P^k$ is always $\vec{w}^T = \pi_*$.

So, $\vec{w}$ is called the **equilibrium** or **stationary** distribution of the chain.

Remark: $\vec{w}^T P = \vec{w}^T$ if and only if $\pi_* P = \pi_*$ if and only if $P^T \vec{w} = \vec{w}$

So, $\vec{w}$ is eigenvector of P^T with eigenvalue 1.

Example 1 Recall the balls in boxes model:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{Solve } P^T \vec{w} = \vec{w}, \text{ we have } \vec{w}^T = \begin{bmatrix} \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{bmatrix}$$

Compute

$$P^{10} = \begin{bmatrix} 0.167 & 0.666 & 0.167 \\ 0.1665 & 0.667 & 0.1665 \\ 0.167 & 0.666 & 0.167 \end{bmatrix}$$

Showing the rate of convergence.

Meaning of Perron-Frobenius

Recall that $\mathbb{P}(X_k = j | X_0 = i) = p_{ij}^{(k)} = (P^k)_{ij}$

So, $\lim_{k \rightarrow \infty} \mathbb{P}(X_k = j | X_0 = i) \rightarrow \pi_j$.

Suppose the initial distribution is $\mathbb{P}(X_0 = i) = \mu_i$

Now, using the law of total probability,

$$\begin{aligned}\mathbb{P}(X_k = j) &= \sum_i \mathbb{P}(X_k = j | X_0 = i) \mathbb{P}(X_0 = i) \\ &= \sum_i (P^k)_{ij} \mu_i \\ &= (\mu^T P^k)_j \\ &= \pi_j\end{aligned}$$

The Theorem says that the state of the chain rapidly forgets its initial distribution and converges to the stationary value.

Irreducible Case of Perron-Frobenius Theorem:

Theorem. Suppose P is an *irreducible* $n \times n$ transition matrix. Then there is a **unique, strictly positive** probability vector $\vec{w} \in \mathbb{R}^n$ such that

$$\vec{w}^T P = \vec{w}^T$$

Furthermore, the Cesàro average converges

$$\frac{1}{n+1} (I + P + P^2 + \dots + P^n) \xrightarrow{\text{red}} W = \begin{bmatrix} - & \vec{w}^T & - \\ - & \vec{w}^T & - \\ & \vdots & \\ - & \vec{w}^T & - \end{bmatrix} = \mathbf{1} \vec{w}^T \quad \text{as } n \rightarrow \infty$$

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Long-run fraction of time

For a Markov chain $\{X_n\}_{n \geq 0}$, define the **empirical time average**

$$\frac{1}{n+1} \sum_{k=0}^n \mathbb{I}_{\{X_k=j\}},$$

which is the fraction of times (up to time n) that the chain is in state j .

$$\frac{1}{n+1} \sum_{k=0}^n \mathbb{I}_{\{X_k=j\}} \rightarrow \pi_j \quad (\text{converge in Expectation}).$$

Reason:

$$E \left[\frac{1}{n+1} \sum_{k=0}^n \mathbb{I}_{\{X_k=j\}} \right] = \frac{1}{n+1} \sum_{k=0}^n E[\mathbb{I}_{\{X_k=j\}}]$$

$$= \frac{1}{n+1} \sum_{k=0}^n \mathbb{P}(X_k = j \mid X_0 = i) = \frac{1}{n+1} \sum_{k=0}^n (P^k)_{ij} \rightarrow \pi_j$$

independent of the starting state i .

Remark: For an irreducible chain, π_j is the long-run fraction of time the chain spends in state j .

A deeper result (the **Ergodic Theorem for Markov Chains**) strengthens the statement:

$$\frac{1}{n+1} \sum_{k=0}^n \mathbf{1}_{\{X_k=j\}} \xrightarrow{a.s.} \pi_j.$$

So not only in expectation, but **along almost every realization**, the fraction of time spent in state j converges to π_j .

Example.

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Example: Drunkard's walk

□ Sojourn times

Theorem (Return times and hitting times)

Let P be the transition matrix of a **finite irreducible Markov chain**.

Let $\pi_* = (\pi_1, \dots, \pi_n)$ be its **stationary distribution**.

Define:

- m_{ij} : Expected number of steps to **hit state j for the first time**, starting from state i (with $m_{jj} = 0$),
- r_i : Expected number of steps to **return to state i** , starting from i .

Then,

$$r_i = \frac{1}{\pi_i} \quad \text{and} \quad m_{ij} = \frac{Z_{jj} - Z_{ij}}{\pi_j}$$

where $Z = (I - P + W)^{-1}$ is the fundamental matrix.

Proof: The proof has **three ideas**:

1. First-step decomposition (probability),
2. Stationarity (Perron–Frobenius),
3. Linear algebra to solve the equations.

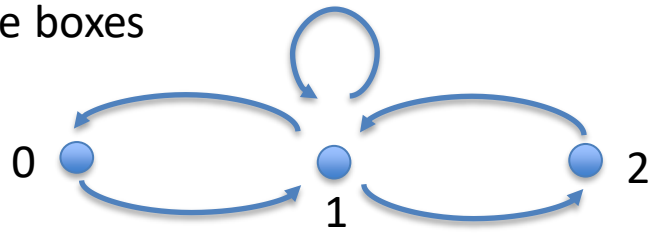
Recall that given $X_0 = i$, the First Return Time:

$$T_i := \inf_n \{n \geq 1 \mid X_n = i\}$$

So, Expected number of steps to **return to state** i , starting from i is

$$r_i := E[T_i]$$

Example: Ball in the boxes



❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross